

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Hospital Clinical NLP — Metropolitan France (DrBenchmark)

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

DrBenchmark's task taxonomy spans 20 tasks across POS, NER, MCQA, MCC, MLC, intent classification, and STS [Q1, Q4, Q18, Q81], covering most of the deployment's required functions: POS tagging (CAS, ESSAI), NER with clinical/temporal/anatomical/treatment entities (DEFT-2021, E3C, QUAERO), and multi-label pathology classification (DEFT-2021 MeSH, DiaMed ICD-10 chapters, MorFITT specialties). However, several ontological gaps directly affect the deployment. First, the elicitation flagged tropical/DOM-TOM pathology as a HIGH-priority future extension; while the deployment scope is Metropolitan France only, DiaMed's source (Pan African Medical Journal) introduces sub-Saharan/North African case content that does not match the metropolitan target population (DIAMED-D7, DIAMED-D8, DIAMED-D9). Second, MorFITT's 12 specialty labels and CAS/ESSAI's negation/speculation labels do not map to ICD-10/PMSI diagnostic axes (MORFITT-D11, CAS-D19, ESSAI-D9), introducing construct-irrelevant variance into aggregate benchmark scores. Third, FrenchMedMCQA's pharmacy-exam content (analytical chemistry, electroanalysis, health economics) is largely irrelevant to hospital clinical NLP (FRENCHMEDMCQA-D2, FRENCHMEDMCQA-D23, FRENCHMEDMCQA-D43, FRENCHMEDMCQA-D45). Fourth, the deployment requires confidence-scored multi-label outputs and uncertainty flagging, but no benchmark task category targets confidence-calibrated or ranked-candidate outputs [Q56]. The taxonomy partially aligns with metropolitan deployment needs but contains both omissions (PMSI-granular codes, calibration tasks) and irrelevant categories (veterinary, pharmacy exam, encyclopedic content).

ID	Evidence
Q1	"It encompasses 20 diversified tasks, including named-entity recognition, part-of-speech tagging, question-answering, semantic textual similarity, and classification." (p.1)
Q4	"These tasks encompass part-of-speech (POS) tagging, named-entity recognition (NER), classification, question-answering (QA), and semantic textual similarity (STS)." (p.1)
Q18	"A variety of tasks with different requirements and objectives: Part-of-Speech (POS) tagging, Multi-class, Multi-label and Intent classification, Named-Entity Recognition (NER), Multiple-Choice Question-Answering (MCQA), and Semantic Textual Similarity (STS)." (p.2)
Q56	"The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions." (p.6)
Q81	"Table 8 summarizes the results obtained on average by the considered MLMs when aggregating the tasks into one of the five designated categories: POS, NER, MCQA, MCC (Multi-class classification), MLC (Multi-label classification), or STS tasks." (p.8)
FRENCHMED MCQA-D2	La loi de Beer-Lambert: S'applique aux dosages spectrophotométriques... Est une loi additive — Analytical chemistry question irrelevant to clinical NLP
DIAMED-D7	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabe hospital origin — non-metropolitan French clinical setting
DIAMED-D8	À propos d'un cas clinique ... keywords: ['Niamey'] — Niger-based case — sub-Saharan African clinical context, not metropolitan France
DIAMED-D9	keywords: ['Rate', 'abdomen aigu', 'scanner abdominal', 'chirurgie', 'Maroc'] — Moroccan clinical case — North African origin rather than French metropolitan
ESSAI-D9	Essai clinique d'immunothérapie évaluant l'association du galunisertib (un facteur de croissance transformant β), avec l'anticorps anti-PD-L1 durvalumab (MEDI4736), dans le cancer du pancréas métastatique — Rich pathology content labeled only for negation/speculation, not ICD-10 — output ontology mismatch
MORFITT-D11	Nous rapportons le cas d'une patiente de 71 ans ayant développé un pseudomyxome péritonéal secondaire à un adénocarcinome appendiculaire. — Clinical case abstract with surgery+etiology labels; labels do not map to ICD-10 codes

CAS-D19	label_names: ["negation_speculation", "negation", "neutral", "speculation"] — Classification labels measure linguistic modality, not ICD-10 pathology categories
FRENCHMED MCQA-D23	En électrophorèse capillaire haute performance, le sens de migration de l'analyse dépend: De la nature de la charge de l'analyte — Analytical chemistry instrumentation; not relevant to clinical NLP
FRENCHMED MCQA-D43	L'économie médicale est une économie... Caractérisée par le libre choix des médecins par le malade — Health economics question; low clinical NLP relevance
FRENCHMED MCQA-D45	Parmi les techniques voltampérométriques, on trouve: La polarographie — Electroanalytical chemistry; irrelevant to hospital clinical NLP

Strengths

- Native multi-label classification structure in DEFT-2021 with up to 7 concurrent MeSH labels per case directly exercises multi-morbid pathology classification (DEFT2021-D1, DEFT2021-D2)
- DEFT-2021 NER scheme covers PATHOLOGY, TREATMENT, ANATOMY, DATE, DURATION, FREQUENCY, MOMENT — full alignment with deployment's clinical and temporal entity extraction needs (DEFT2021-D3, DEFT2021-D4, DEFT2021-D5)
- Breadth of 20 tasks ensures most deployment functions (POS, NER, MLC, temporal extraction) have at least one structurally aligned dataset

ID	Evidence
DEFT2021-D1	Un homme de 42 ans était hospitalisé en urgence le 10 décembre 2003 dans le service de gastro-entérologie pour un tableau d'hépatite aiguë — Multi-morbid hepatology/toxicology/psychiatry case with 6 concurrent MeSH labels
DEFT2021-D2	Nous rapportons le cas d'un garçon de huit ans [...] microangiopathie thrombotique secondaire à un syndrome hémolytique urémique (SHU) atypique — 7-label pediatric multi-morbid case illustrating complex label overlap
DEFT2021-D3	cotrimoxazole pour 10 jours — Treatment drug with duration span annotated — confirms temporal treatment labeling
DEFT2021-D4	Hydrocortisone 300 mg IV immédiatement ; Traitement Hydroxyzine 25 mg par voie orale toutes les 6 heures — Multiple entity types co-occurring in a single sentence, confirming NER scheme coverage
DEFT2021-D5	Au jour 42 , une deuxième désensibilisation a eu lieu — Temporal anchor (relative day) labeled, confirming temporal entity extraction alignment

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required deployment categories include: POS tagging on clinical text, clinical-entity NER (pathology, anatomy, treatment, dosage), temporal-entity NER (date, duration, frequency, moment), multi-label pathology classification mapped to ICD-10/PMSI, and confidence-scored ranked-candidate output. DrBenchmark covers POS (CAS, ESSAI) [Q42, Q43], clinical NER [Q108], temporal NER [Q104, Q108, DEFT2021-D3, DEFT2021-D5], and multi-label classification [Q26, Q27, Q45, DEFT2021-D1].

IO-2: Confidence-scored / ranked-candidate / uncertainty-flagged output categories are absent — the benchmark uses standard classification metrics and majority-class baselines [Q56]. PMSI-granular code classification is also absent; DiaMed only reaches ICD-10 chapter level (DIAMED-D11). Tropical/DOM-TOM pathology categories not present in French clinical subsets (E3C-D4, E3C-D13).

IO-3: FrenchMedMCQA pharmacy exam content covering electroanalysis, analytical chemistry, and health economics is largely irrelevant to clinical NLP deployment (FRENCHMEDMCQA-D23, FRENCHMEDMCQA-D43, FRENCHMEDMCQA-D45). MorFITT contains substantial veterinary content (equine, porcine, canine) off-domain for human hospital deployment (MORFITT-D5, MORFITT-D9, MORFITT-D26). DEFT-2020 contains 30–40% off-domain encyclopedic content (railways, video games, sports) (DEFT2020-D9, DEFT2020-D10, DEFT2020-D11, DEFT2020-D12). CAS/ESSAI negation/speculation classification labels are linguistically rather than clinically relevant to ICD-10 mapping (CAS-D19, ESSAI-D9).

IO-4: Construct underrepresentation: confidence calibration, PMSI-granular codes, DOM-TOM tropical pathology (out of immediate scope but flagged priority extension). Construct-irrelevant variance: pharmacy exam content, veterinary content, encyclopedic STS pairs. Aggregate scores across 20 tasks may obscure deployment-relevant performance.

ID	Evidence
Q1	"It encompasses 20 diversified tasks, including named-entity recognition, part-of-speech tagging, question-answering, semantic textual similarity, and classification." (p.1)
Q18	"A variety of tasks with different requirements and objectives: Part-of-Speech (POS) tagging, Multi-class, Multi-label and Intent classification, Named-Entity Recognition (NER), Multiple-Choice Question-Answering (MCQA), and Semantic Textual Similarity (STS)." (p.2)
Q26	"The multi-label classification task focuses on identifying the patient's clinical profile based on the diseases, signs, or symptoms mentioned in the clinical cases." (p.3)
Q27	"The dataset is annotated with 23 axes from Chapter C of the Medical Subject Headings (MeSH)." (p.3)
Q39	"FrenchMedMCQA (Labrak et al., 2022) is a Multiple-Choice Question-Answering (MCQA) dataset for biomedical domain. It contains 3,105 questions coming from real exams of the French medical specialization diploma in pharmacy, integrating single and multiple answers. The first task consists of automatically identifying the set of correct answers among the 5 proposed for a given question. The second task consists of identifying the number of answers (between 1 and 5) supposedly correct for a given question." (p.4)
Q42	"CAS (Grabar et al., 2018) comprises 3,790 clinical cases that have been annotated for POS tagging with 31 classes using automatic annotations through Tagex 3, with an evaluation conducted by comparing the automatic outputs against manual annotations. This evaluation yielded 98% precision. Since the dataset does not come with predefined subsets, we made the decision to randomly split it into 3 subsets of 70%, 10% and 20% of the total data for training, validation and test respectively." (p.4)
Q43	"ESSAI (Dalloux et al., 2021) contains 7,247 clinical trial protocols annotated in 41 POS tags using TreeTagger (Schmid, 1994). As the dataset was not originally divided into 3 subsets, we applied the same procedure as on the CAS corpus." (p.4)
Q45	"DiaMed is an original dataset created specifically for DrBenchmark. It comprises 739 new French clinical cases collected from an open source journal (The Pan African Medical Journal). The cases have been manually annotated by several annotators, one of which is a medical expert, into 22 chapters of the International Classification of Diseases, 10th Revision (ICD-10) (Wor, 2019). These chapters provide a general description of the type of injury or disease. To ease the annotation process, only label at the chapter level were used (more detail in Appendix B.8). The inter-annotator agreement between the 4 annotators has been computed for two annotation sessions (see Table 4), with 15 different clinical cases assessed per session." (p.4)
Q56	"The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions." (p.6)
Q104	"Clinical: O, and CLINENTITY. Temporal: O, EVENT, ACTOR, BODYPART, TIMEX3 and RML" (p.14)
Q108	"Named-entity recognition: O, ANATOMY, DATE, DOSAGE, DURATION, MEDICAL EXAM, FREQUENCY, MODE, MOMENT, PATHOLOGY, SOSY, SUBSTANCE, TREATMENT and VALUE" (p.14)
DEFT2021-D1	Un homme de 42 ans était hospitalisé en urgence le 10 décembre 2003 dans le service de gastro-entérologie pour un tableau d'hépatite aiguë — Multi-morbid hepatology/toxicology/psychiatry case with 6 concurrent MeSH labels
DEFT2021-D3	cotrimoxazole pour 10 jours — Treatment drug with duration span annotated — confirms temporal treatment labeling
E3C-D4	The presence of terminal spine confirmed Schistosoma haematobium and a diagnosis of schistosomiasis was made. — Tropical parasitic pathology in English config — confirms tropical content structurally absent from French subset
DEFT2021-D5	Au jour 42 , une deuxième désensibilisation a eu lieu — Temporal anchor (relative day) labeled, confirming temporal entity extraction alignment
MORFITT-D5	seules des données limitées sont disponibles sur le surfactant chez le cheval... notre objectif était de mener une étude immunocytochimique au microscope optique et électronique sur les poumons normaux du cheval. — Equine veterinary immunology study; off-domain for human hospital deployment
MORFITT-D9	L'otite externe est une maladie multifactorielle fréquente chez le chien... Décrire le microbiote bactérien auriculaire des chiens avec otite externe comparé aux chiens sains. — Canine ear microbiome study; veterinary content off-domain for hospital NLP

DEFT2020-D9	Entre Perpignan et Villefranche, il subsiste de très nombreux poteaux caténaux datant des premiers essais en 12 KV 16 2/3 Hz... — Railway infrastructure content; entirely off-domain for clinical NLP deployment
ESSAI-D9	Essai clinique d'immunothérapie évaluant l'association du galunisertib (un facteur de croissance transformant β), avec l'anticorps anti-PD-L1 durvalumab (MEDI4736), dans le cancer du pancréas métastatique — Rich pathology content labeled only for negation/speculation, not ICD-10 — output ontology mismatch
DEFT2020-D10	Boris Fiodorovitch Godounov, en russe : Борис Фёдорович Годунов (v.1551–Moscou, 13 avril 1605)... — Russian historical figure biography; off-domain and contains Cyrillic script
DIAMED-D11	Maladie de Kaposi à localisation broncho-pulmonaire révélant une infection VIH — Chapter-level label only; PMSI granular codes (e.g., B21.x) not evaluated
DEFT2020-D11	Les Canadiens de Montréal sont une franchise de hockey sur glace professionnel située à Montréal dans la province de Québec au Canada. — Professional hockey franchise; irrelevant to hospital deployment
DEFT2020-D12	Lancement du célèbre MMORPG : World of Warcraft. — Video game launch event; entirely off-domain
E3C-D13	a diagnosis of schistosomiasis was made — Tropical disease in English config only; absent from French subset — confirms DOM-TOM pathology gap
CAS-D19	label_names: ["negation_speculation", "negation", "neutral", "speculation"] — Classification labels measure linguistic modality, not ICD-10 pathology categories
FRENCHMED MCQA-D23	En électrophorèse capillaire haute performance, le sens de migration de l'analyse dépend: De la nature de la charge de l'analyte — Analytical chemistry instrumentation; not relevant to clinical NLP
MORFITT-D26	Prévalence et caractéristiques morphologiques et moléculaires de Sarcocystis bertrami chez les chevaux en Chine. — Horse parasitology study from China; off-domain for human hospital deployment
FRENCHMED MCQA-D43	L'économie médicale est une économie... Caractérisée par le libre choix des médecins par le malade — Health economics question; low clinical NLP relevance
FRENCHMED MCQA-D45	Parmi les techniques voltampérométriques, on trouve: La polarographie — Electroanalytical chemistry; irrelevant to hospital clinical NLP
WEB-2	PMSI uses GHM groupings derived from CIM-10 FR principal/secondary diagnoses + CCAM procedures + age + length of stay — substantially more granular than ICD-10 chapter labels (https://www.lespmsi.com/2024/manuel_GHM_2024_volume1_provisoire.pdf)

Information Gaps

- Whether aggregate benchmark scores can be decomposed to filter out off-domain content (veterinary, encyclopedic, pharmacy exam) for deployment-relevant interpretation

Requires Expert Verification

- Whether DEFT-2021's 23 MeSH axes provide adequate proxy for PMSI Catégorie Majeure groupings used in operational hospital coding

Remediation

Gap 1: Several DrBenchmark tasks contribute construct-irrelevant variance to aggregate scores — FrenchMedMCQA pharmacy exam (chemistry/economics), MorFITT veterinary content, DEFT-2020 encyclopedic pairs

Recommendation: When interpreting DrBenchmark performance for this deployment, decompose aggregate scores task-by-task and weight DEFT-2021 (NER + MLC), CAS (POS + clinical case register), CLISTER (clinical STS), and E3C French_temporal as deployment-relevant; treat FrenchMedMCQA, MorFITT veterinary subsets, and DEFT-2020 encyclopedic content as low-relevance signal for aggregate interpretation.